



មហាវិទ្យាល័យវិស្វកម្ម
FACULTY OF ENGINEERING

Computer Programing

Lecture 1

Introduction to Computer Programming with C++

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Contents

- Course orientation CP
- Programming Languages
- Code Editor
- Operating Systems
- Some of the key features of programming languages

Course orientation CP

Objective:

- The central theme of this course is to learn how to solve problems by writing a program.

**Problem Solving
in Computer
Programming**

**Problem Solving, Logical
Thinking, How to Solve
Exam Problems?**

Course orientation CP

Syllabus

Week	Lecture
1	Introduction to Computer Programming
2	Getting Started with C++
3	Values and Variables - 1
4	Values and Variables - 2
5	Conditional Execution
6	Iteration: Loop/while/for – 1
7	Iteration: Loop/while/for – 2
8	Midterm
9	Using Functions – Built-in functions
10	Writing Functions – User defines the function – 1
11	Writing Functions – User defines the function – 2
12	Class – Object - 1
13	Class – Object - 2
14	Array
15	Review
16	Final

Course orientation CP

Assessments

- Attendance: 10%
- Group Discussion: 10%
- Independent Homework: 20%
- Midterm 20%
- Final Exam: 40%

Programming Languages

- A **programming language** is a formal system of communication used by programmers to write instructions that a computer can understand and execute.



C#



C++



TypeScript



Golang



Rust

Programming Languages

Types of Programming Languages

- Low-Level Languages
 - Machine Language (Binary): Direct instructions to hardware (e.g., 10101010)
 - Assembly Language: Symbolic representation of machine instructions Example: MOV AX, BX

Programming Languages

Types of Programming Languages

- High-Level Languages
 - Closer to human language; easier to read/write
- Example: C, C++, Python, Java, JavaScript

Programming Languages

Types of Programming Languages

- Very High-Level / Domain-Specific Languages
 - Tailored for specific tasks or domains
 - Example: SQL (databases), HTML/CSS (web), MATLAB (engineering)

Programming Languages

Popular Programming Languages & Uses

Language	Use Cases
Python	AI, data science, automation, web
C++	Games, systems, performance apps
Java	Enterprise, Android, cross-platform
JavaScript	Web development (frontend/backend)
C	Operating systems, embedded systems
Go	Cloud computing, fast web servers
Rust	Safe systems programming
Swift	iOS/macOS app development
Kotlin	Modern Android development

Code Editor

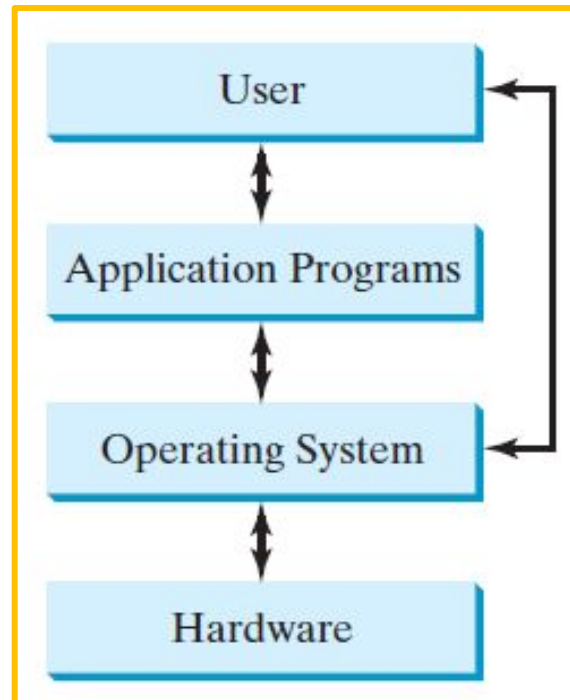
- A **code editor** is a tool that developers use to help them write code more efficiently
- Where the developer can enter their code
- Color-coded syntax highlighting, automatic indentation, error checking, auto-complete with suggested code

Code Editor

1. Visual Studio Code
2. Sublime Text
3. PyCharm from JetBrains
4. ATOM
5. IntelliJ IDEA
6. Vim
7. Espresso
8. Apache NetBeans
9. Notepad++
10. Brackets
11. Code Blocks

Operating Systems

- The operating system (OS) is the most important program that runs on a computer.
- The OS manages and controls a computer's activities.



Operating Systems

Examples of Operating Systems

- **Windows** (GUI-based, PC) GNU/Linux (Personal, Workstations, ISP, File, and print server, Three-tier client/Server)
- **macOS** (Macintosh), used for Apple's personal computers and workstations (MacBook, iMac).
- **Android** (Google's Operating System for smartphones/tablets/smartwatches)
- **iOS** (Apple's OS for iPhone, iPad, and iPod Touch)

Some of the key features of programming languages

- Syntax
- Data Types
- Variables
- Operators
- Control Structures
- Function
- Libraries
- Class and object

CP – W1 – Practice

Install C++ with Visual Studio Code

Step1: Download and Install Mingw

<https://www.youtube.com/watch?v=0HD0pqVtsmw&t=0s>

Step 2: How to Download and Install Visual Studio Code (VS Code) on Windows

<https://www.youtube.com/watch?v=JGsyJI8XG0Y&t=0s>

Step 3: How to Set up Visual Studio Code for C and C++ Programming

https://www.youtube.com/watch?v=77v-Poud_io

Install C++ with

1. Visual Studio Code
2. Code Blocks
3.
4. c++ online compiler
 - OnlineGDB
 - Link:
https://www.onlinegdb.com/online_c++_compiler

Ex 1. Getting Start C++

- Capture the features of your program after successful installation
- Save the field *name.cpp*
- Get started with simple

A Simple C++ Program

- A C++ program is executed from the main function.

include library	1	<code>#include <iostream></code>
using namespace	2	<code>using namespace std;</code>
	3	
main function	4	<code>int main()</code>
	5	<code>{</code>
comment	6	<code>// Display Welcome to C++ to the console</code>
output	7	<code>cout << "Welcome to C++!" << endl;</code>
	8	
successful return	9	<code>return 0;</code>
	10	<code>}</code>

Programming Style and Documentation

- Good programming style and proper documentation make a program easy to read and help programmers prevent errors.

```
cout << 3+4*4;
```

← Bad style

```
cout << 3 + 4 * 4;
```

← Good style

Programming Errors

- Programming errors can be categorized into three types:
 - syntax errors,
 - runtime errors,
 - logic errors.

Programming Errors

- Syntax Errors

```
1  #include <iostream>
2  using namespace std
3
4  int main()
5  {
6      cout << "Programming is fun << endl;
7
8      return 0;
9  }
```

Compile

→ 1>Test.cpp(4): error C2144: syntax error : 'int' should be preceded by ';'
1>Test.cpp(6): error C2001: newline in constant
1>Test.cpp(8): error C2143: syntax error : missing ';' before 'return'

Programming Errors

- Runtime Errors

runtime error

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int i = 4;
7      int j = 0;
8      cout << i / j << endl;
9
10     return 0;
11 }
```


Programming Errors

- Logic Errors

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      cout << "Celsius 35 is Fahrenheit degree " << endl;
7      cout << (9 / 5) * 35 + 32 << endl;
8
9      return 0;
10 }
```

```
Celsius 35 is Fahrenheit degree
67
```

Common Errors

- **Missing Braces**

```
int main()
```

```
{
```

```
}
```

← Type this closing brace right away to match the opening brace

- **Missing Semicolons**

```
int main()
```

```
{
```

```
    cout << "Programming is fun!" << endl;
```

```
    cout << "Fundamentals First" << endl;
```

```
    cout << "Problem Driven" << endl
```

```
}
```

Missing a semicolon

Common Errors

- **Missing Quotation Marks**

```
cout << "Problem Driven;
```

↑
Missing a quotation mark

- **Misspelling Names**

```
1 int Main()  
2 {  
3     cout << (10.5 + 2 * 3) / (45 - 3.5) << endl;  
4     return 0;  
5 }
```

Thanks!